Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A ball is thrown vertically upward at 15 m/s from h=0. Use g=10 and neglect drag. Find maximum height by energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mg Δh = $\Delta K_{translation}$ + $\Delta K_{rotation}$ + $\Delta E_{thermal}$. Closed energy accounting near Earth; Δh is the positive drop. Determine whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses 6 J of gravitational potential energy but gains only 4 J of translational kinetic energy. The 2 J difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the 2 J remainder explicitly labeled as not yet identified.

Initial **Transfers** **Final**

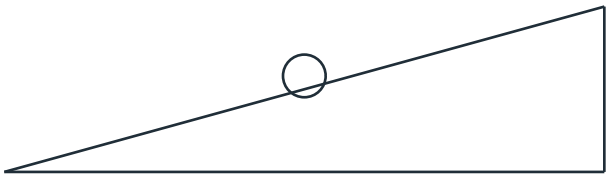


Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

04 Apply the physics

A lab report says a 0.50 kg cart starts from rest, drops 0.20 m vertically and reaches 2.5 m/s. Use g = 9.8 N/kg. Compare the gravitational energy available with the reported final kinetic energy. What should the group check before claiming energy was created?

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Create a table for repeated trials. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed momentum magnitude, mass becomes 4 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Supplied data: m=0.50 kg, h=0.40 m, v=2.4 m/s, g=9.8. Find mgh, K and fraction of initial gravitational energy appearing as translation. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg cart drops 0.4 m vertically and reaches 2 m/s from rest. Use g=10. Find mechanical energy not in translation and average resistive force if all of it is friction over a 1 m path. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3